

UNIT 3-Hands on Descriptive Statistics

Name: K. L. Khaviya varshini

Reg. No.: 192472381

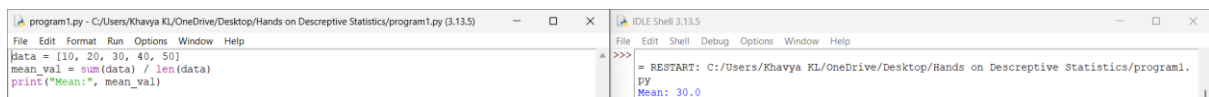
PROGRAM 1:

```
data = [10, 20, 30, 40, 50]

mean_val = sum(data) / len(data)

print("Mean:", mean_val)
```

OUTPUT:

The screenshot shows a Python IDE with two windows. The left window, titled 'program1.py', contains the code for Program 1: data = [10, 20, 30, 40, 50], mean_val = sum(data) / len(data), and print("Mean:", mean_val). The right window, titled 'IDLE Shell 3.13.5', shows the execution output: >>> = RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.py, Mean: 30.0.

PROGRAM 2:

```
data = [40, 10, 30, 50, 20]

data.sort()

n = len(data)

if n % 2 == 0:

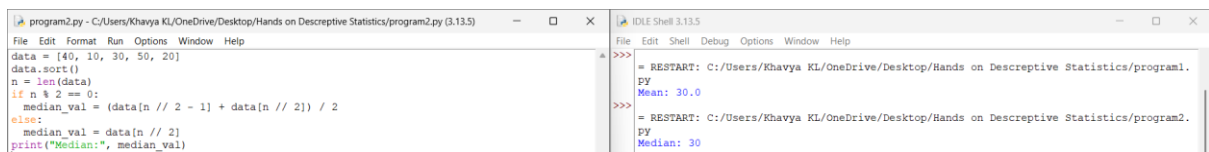
    median_val = (data[n // 2 - 1] + data[n // 2]) / 2

else:

    median_val = data[n // 2]

print("Median:", median_val)
```

OUTPUT:

The screenshot shows a Python IDE with two windows. The left window, titled 'program2.py', contains the code for Program 2: data = [40, 10, 30, 50, 20], data.sort(), n = len(data), if n % 2 == 0: median_val = (data[n // 2 - 1] + data[n // 2]) / 2, else: median_val = data[n // 2], and print("Median:", median_val). The right window, titled 'IDLE Shell 3.13.5', shows the execution output: >>> = RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.py, Mean: 30.0, >>> = RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program2.py, Median: 30.

PROGRAM 3:

```
from collections import Counter

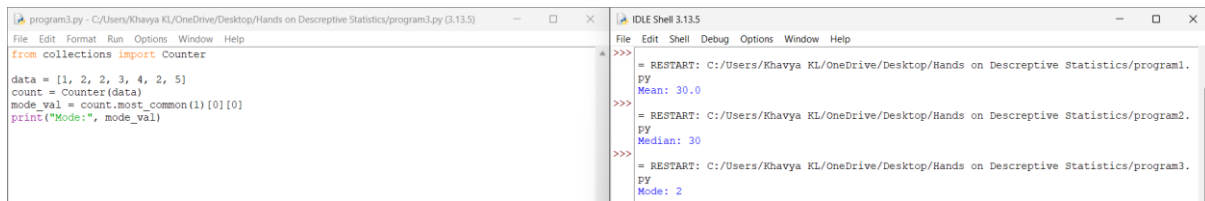
data = [1, 2, 2, 3, 4, 2, 5]

count = Counter(data)

mode_val = count.most_common(1)[0][0]

print("Mode:", mode_val)
```

OUTPUT:



The screenshot shows two windows. The left window, titled 'program3.py - C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program3.py (3.13.5)', contains the following code:

```
from collections import Counter
data = [1, 2, 2, 3, 4, 2, 5]
count = Counter(data)
mode_val = count.most_common(1)[0][0]
print("Mode:", mode_val)
```

The right window, titled 'IDLE Shell 3.13.5', shows the output of the program:

```
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.
PY
Mean: 30.0
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program2.
PY
Median: 30
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program3.
PY
Mode: 2
```

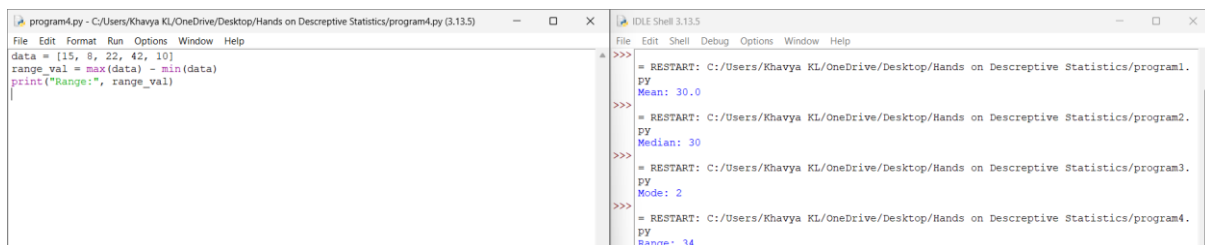
PROGRAM4:

```
data = [15, 8, 22, 42, 10]
```

```
range_val = max(data) - min(data)
```

```
print("Range:", range_val)
```

OUTPUT:



The screenshot shows two windows. The left window, titled 'program4.py - C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.py (3.13.5)', contains the following code:

```
data = [15, 8, 22, 42, 10]
range_val = max(data) - min(data)
print("Range:", range_val)
```

The right window, titled 'IDLE Shell 3.13.5', shows the output of the program:

```
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.
PY
Mean: 30.0
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program2.
PY
Median: 30
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program3.
PY
Mode: 2
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.
PY
Range: 34
```

PROGRAM 5:

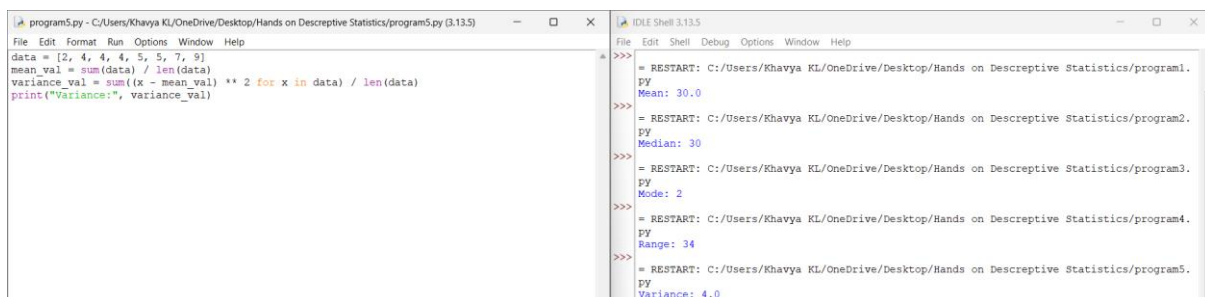
```
data = [2, 4, 4, 4, 5, 5, 7, 9]
```

```
mean_val = sum(data) / len(data)
```

```
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
```

```
print("Variance:", variance_val)
```

OUTPUT:



The screenshot shows two windows. The left window, titled 'program5.py - C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.py (3.13.5)', contains the following code:

```
data = [2, 4, 4, 4, 5, 5, 7, 9]
mean_val = sum(data) / len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
print("Variance:", variance_val)
```

The right window, titled 'IDLE Shell 3.13.5', shows the output of the program:

```
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.
PY
Mean: 30.0
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program2.
PY
Median: 30
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program3.
PY
Mode: 2
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.
PY
Range: 34
>>> = RESTART: C:/Users/Khavva KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.
PY
Variance: 4.0
```

PROGRAM 6:

```
data = [2, 4, 4, 4, 5, 5, 7, 9]
```

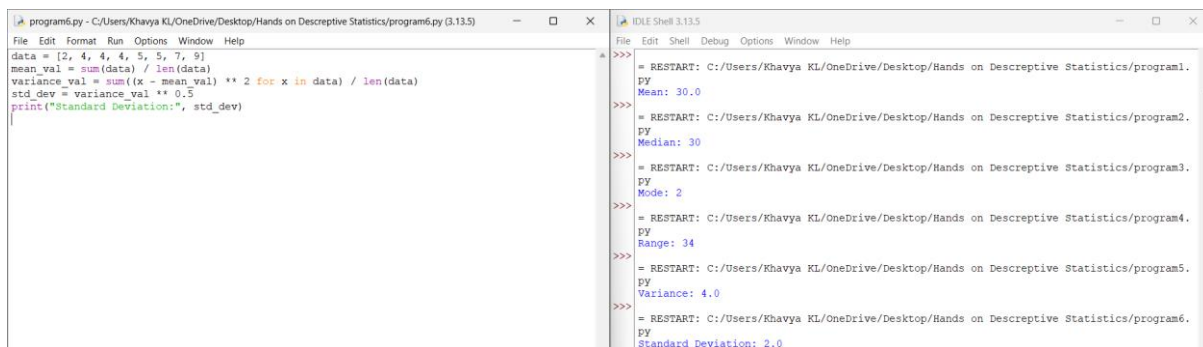
```
mean_val = sum(data) / len(data)
```

```
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
```

```
std_dev = variance_val ** 0.5
```

```
print("Standard Deviation:", std_dev)
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for Program 6, and the right window shows the output. The code calculates the standard deviation of the list [2, 4, 4, 4, 5, 5, 7, 9]. The output shows the standard deviation as 2.0.

```
program6.py - C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program6.py (3.13.5)
File Edit Format Run Options Window Help
data = [2, 4, 4, 4, 5, 5, 7, 9]
mean_val = sum(data) / len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
std_dev = variance_val ** 0.5
print("Standard Deviation:", std_dev)

IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.
PY
Mean: 30.0
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program2.
PY
Median: 30
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program3.
PY
Mode: 2
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.
PY
Range: 34
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.
PY
Variance: 4.0
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program6.
PY
Standard Deviation: 2.0
```

PROGRAM 7:

```
import statistics
```

```
data = [12, 15, 15, 18, 22, 25]
```

```
print("Mean:", statistics.mean(data))
```

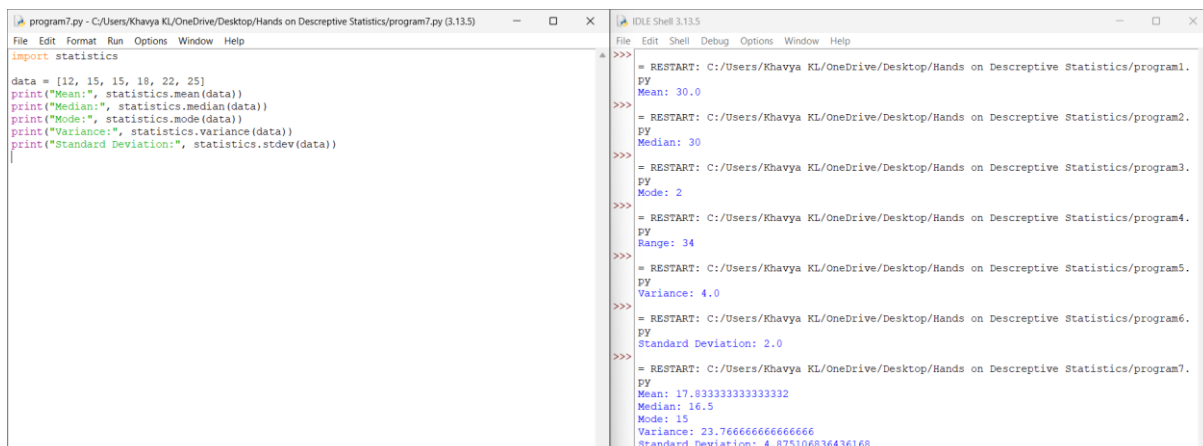
```
print("Median:", statistics.median(data))
```

```
print("Mode:", statistics.mode(data))
```

```
print("Variance:", statistics.variance(data))
```

```
print("Standard Deviation:", statistics.stdev(data))
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for Program 7, and the right window shows the output. The code uses the statistics module to calculate the mean, median, mode, variance, and standard deviation of the list [12, 15, 15, 18, 22, 25]. The output shows the mean as 17.5, median as 16.5, mode as 15, variance as 23.766666666666666, and standard deviation as 4.875106836436168.

```
program7.py - C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program7.py (3.13.5)
File Edit Format Run Options Window Help
import statistics
data = [12, 15, 15, 18, 22, 25]
print("Mean:", statistics.mean(data))
print("Median:", statistics.median(data))
print("Mode:", statistics.mode(data))
print("Variance:", statistics.variance(data))
print("Standard Deviation:", statistics.stdev(data))

IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program1.
PY
Mean: 30.0
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program2.
PY
Median: 30
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program3.
PY
Mode: 2
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.
PY
Range: 34
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.
PY
Variance: 4.0
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program6.
PY
Standard Deviation: 2.0
>>>
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program7.
PY
Mean: 17.5
Median: 16.5
Mode: 15
Variance: 23.766666666666666
Standard Deviation: 4.875106836436168
```

PROGRAM 8:

```
import numpy as np

data = np.array([10, 20, 20, 30, 40])

print("Mean:", np.mean(data))

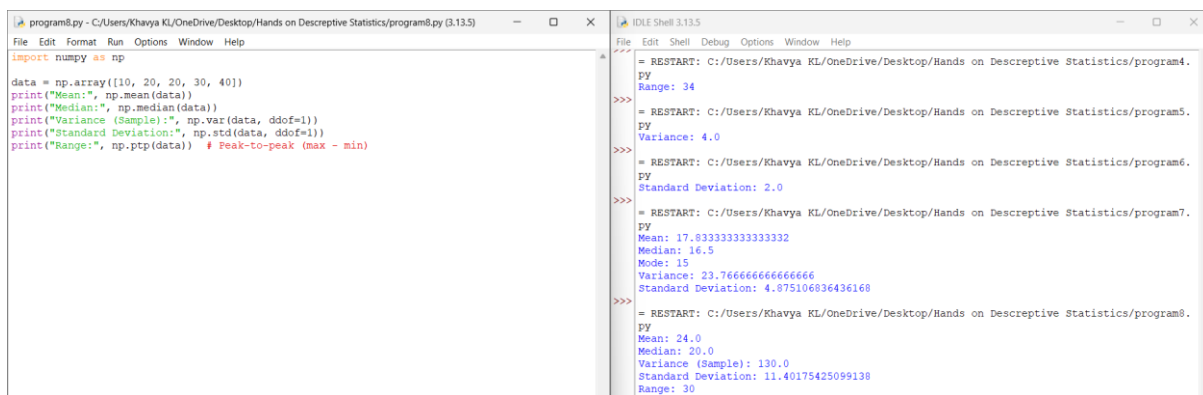
print("Median:", np.median(data))

print("Variance (Sample):", np.var(data, ddof=1))

print("Standard Deviation:", np.std(data, ddof=1))

print("Range:", np.ptp(data)) # Peak-to-peak (max - min)
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for Program 8, and the right window shows the output of the program. The code calculates the mean, median, variance, standard deviation, and range of the array [10, 20, 20, 30, 40]. The output shows the results of these calculations.

```
program8.py - C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program8.py (3.13.5)
File Edit Format Run Options Window Help
import numpy as np
data = np.array([10, 20, 20, 30, 40])
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Variance (Sample):", np.var(data, ddof=1))
print("Standard Deviation:", np.std(data, ddof=1))
print("Range:", np.ptp(data)) # Peak-to-peak (max - min)

IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.py
Range: 34
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.py
Variance: 4.0
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program6.py
Standard Deviation: 2.0
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program7.py
Mean: 17.833333333333332
Median: 16.5
Mode: 15
Variance: 23.766666666666666
Standard Deviation: 4.875106836436168
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program8.py
Mean: 24.0
Median: 20.0
Variance (Sample): 130.0
Standard Deviation: 11.40175425099138
Range: 30
```

PROGRAM 9:

```
from scipy import stats

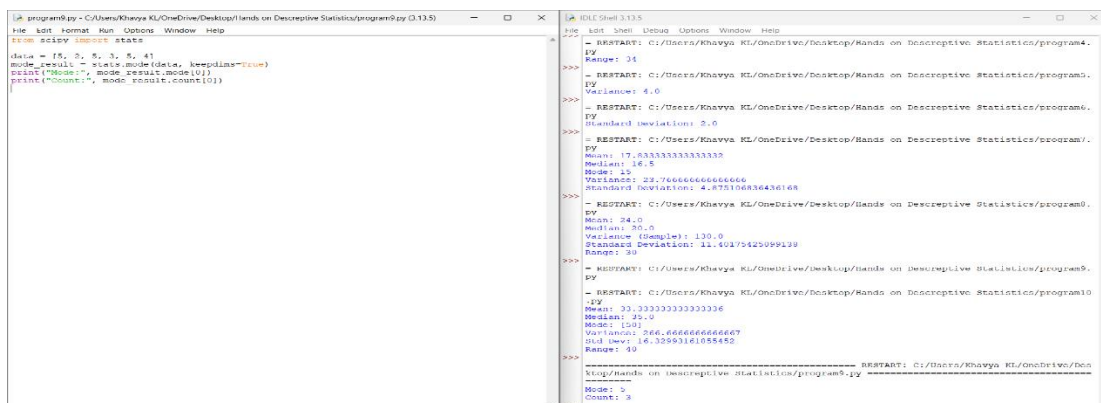
data = [5, 2, 5, 3, 5, 4]

mode_result = stats.mode(data, keepdims=True)

print("Mode:", mode_result.mode[0])

print("Count:", mode_result.count[0])
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for Program 9, and the right window shows the output of the program. The code uses SciPy's mode function to find the mode and count of the array [5, 2, 5, 3, 5, 4]. The output shows the mode and count.

```
program9.py - C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program9.py (3.13.5)
File Edit Format Run Options Window Help
from scipy import stats
data = [5, 2, 5, 3, 5, 4]
mode_result = stats.mode(data, keepdims=True)
print("Mode:", mode_result.mode[0])
print("Count:", mode_result.count[0])

IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.py
Range: 34
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.py
Variance: 4.0
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program6.py
Standard Deviation: 2.0
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program7.py
Mean: 17.833333333333332
Median: 16.5
Mode: 15
Variance: 23.766666666666666
Standard Deviation: 4.875106836436168
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program8.py
Mean: 24.0
Median: 20.0
Variance (Sample): 130.0
Standard Deviation: 11.40175425099138
Range: 30
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program9.py
Mode: 5
Count: 3
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program10.py
Mean: 33.333333333333336
Median: 30.0
Mode: 150
Variance: 366.6666666666667
Standard Deviation: 19.14645503079165
Range: 40
>>> ===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program9.py =====
Mode: 5
Count: 3
```

PROGRAM 10:

```
import pandas as pd

df = pd.DataFrame({'values': [10, 20, 30, 40, 50, 50]})

print("Mean:", df['values'].mean())

print("Median:", df['values'].median())

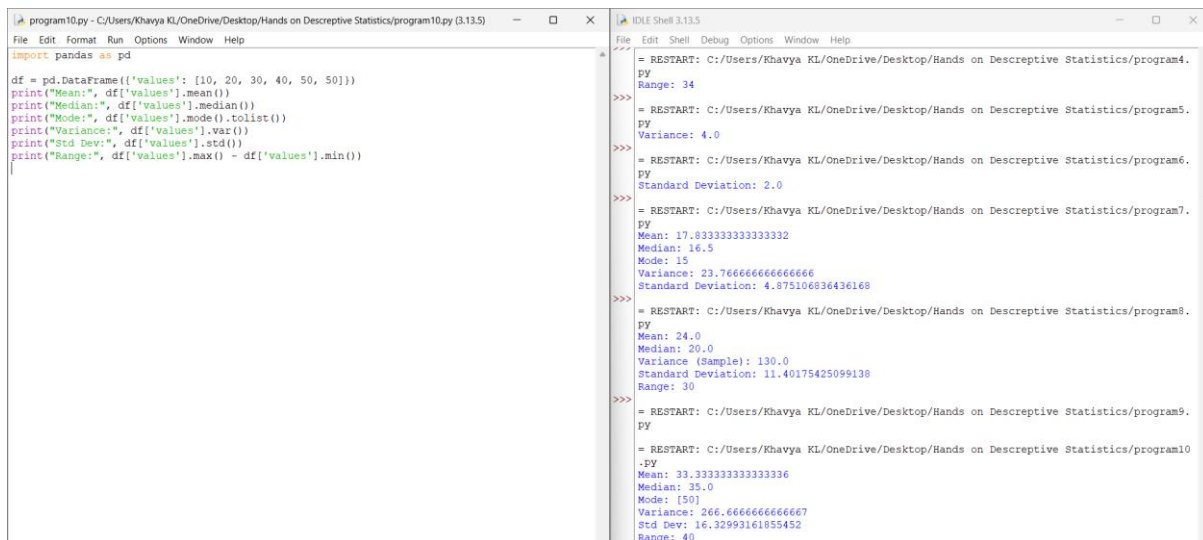
print("Mode:", df['values'].mode().tolist())

print("Variance:", df['values'].var())

print("Std Dev:", df['values'].std())

print("Range:", df['values'].max() - df['values'].min())
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for Program 10, and the right window shows the output of the program.

```
program10.py - C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program10.py (3.13.5)
File Edit Format Run Options Window Help
import pandas as pd

df = pd.DataFrame({'values': [10, 20, 30, 40, 50, 50]})
print("Mean:", df['values'].mean())
print("Median:", df['values'].median())
print("Mode:", df['values'].mode().tolist())
print("Variance:", df['values'].var())
print("Std Dev:", df['values'].std())
print("Range:", df['values'].max() - df['values'].min())
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program4.
.PY
Range: 34
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program5.
.PY
Variance: 4.0
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program6.
.PY
Standard Deviation: 2.0
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program7.
.PY
Mean: 17.833333333333332
Median: 16.5
Mode: 15
Variance: 23.766666666666666
Standard Deviation: 4.875106836436168
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program8.
.PY
Mean: 24.0
Median: 20.0
Variance (Sample): 130.0
Standard Deviation: 11.40175425099138
Range: 30
>>> = RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program9.
.PY
= RESTART: C:/Users/Khavya KL/OneDrive/Desktop/Hands on Descriptive Statistics/program10
.PY
Mean: 33.333333333333336
Median: 35.0
Mode: [50]
Variance: 266.6666666666667
Std Dev: 16.32993161855452
Range: 40
```